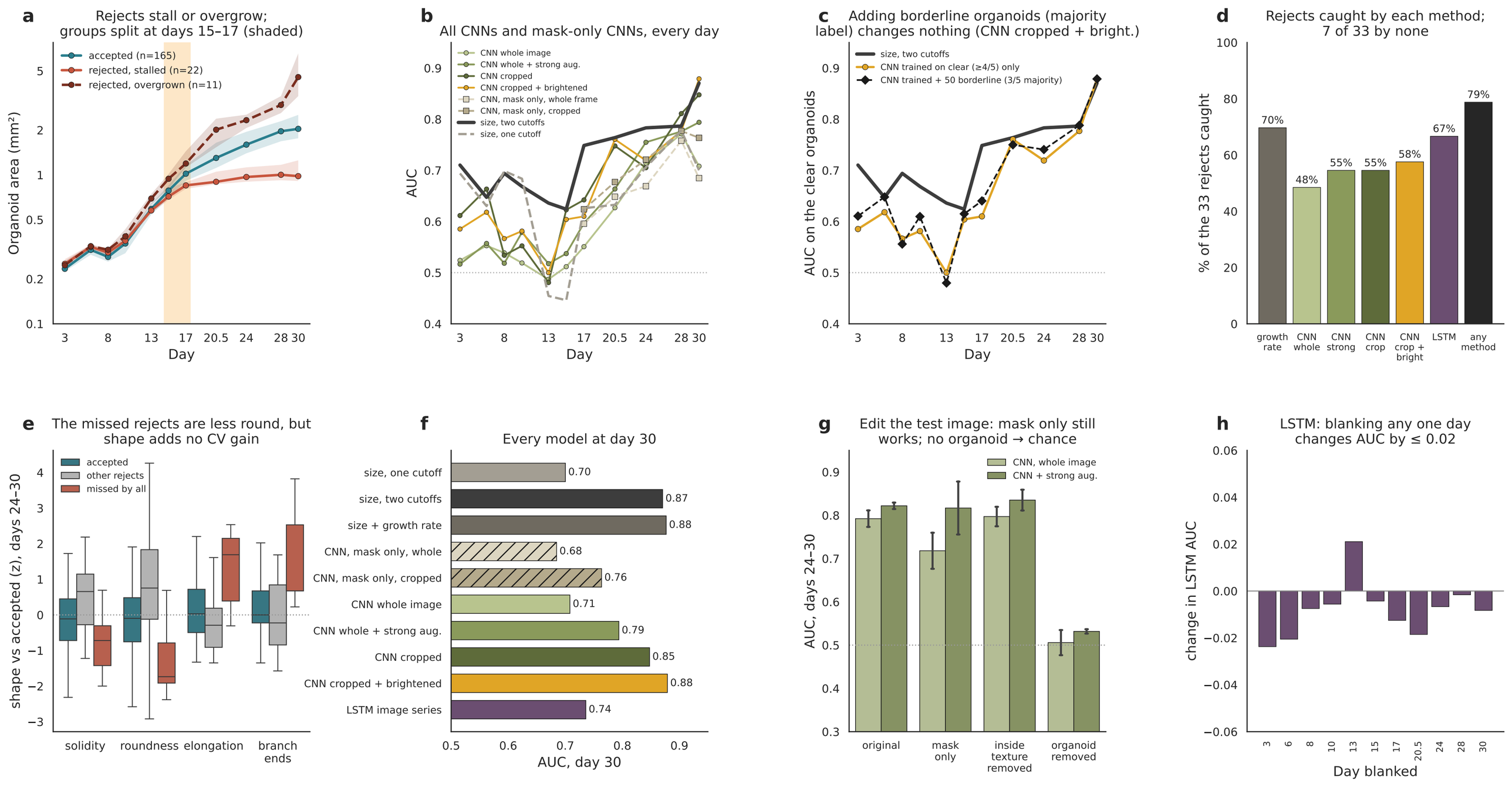
a
Rejects stall or overgrow;
groups split at days 15–17 (shaded)
accepted (n=165)
rejected, stalled (n=22)
rejected, overgrown (n=11)
Organoid area (mm²)
Day
b
All CNNs and mask-only CNNs, every day
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
CNN, mask only, whole frame
CNN, mask only, cropped
size, two cutoffs
size, one cutoff
AUC
Day
c
Adding borderline organoids (majority
label) changes nothing (CNN cropped + bright.)
size, two cutoffs
CNN trained on clear (≥4/5) only
CNN trained + 50 borderline (3/5 majority)
AUC on the clear organoids
Day
d
Rejects caught by each method;
7 of 33 by none
% of the 33 rejects caught
70%
48%
55%
55%
58%
67%
79%
growth rate
CNN whole
CNN strong
CNN crop
CNN crop + bright
LSTM
any method
e
The missed rejects are less round, but
shape adds no CV gain
accepted
other rejects
missed by all
shape vs accepted (z), days 24–30
solidity
roundness
elongation
branch ends
f
Every model at day 30
size, one cutoff 0.70
size, two cutoffs 0.87
size + growth rate 0.88
CNN, mask only, whole 0.68
CNN, mask only, cropped 0.76
CNN whole image 0.71
CNN whole + strong aug. 0.79
CNN cropped 0.85
CNN cropped + brightened 0.88
LSTM image series 0.74
AUC, day 30
g
Edit the test image: mask only still
works; no organoid → chance
CNN, whole image
CNN + strong aug.
AUC, days 24–30
original
mask only
inside texture removed
organoid removed
h
LSTM: blanking any one day
changes AUC by ≤ 0.02
change in LSTM AUC
Day blanked